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0.1 Question 2 (4 pts)

Let A and B be events in a sample space S .

Suppose that the probability that A occurs is 0.3, the probability that B occurs is 0.4, and the probability that neither A nor B occur is 0.5 (that is, the joint probability that A doesn't occur AND B doesn't occur is 0.5).

Determine if you have enough informatin to answer the following questions. If so, write your solution showing all steps. If not, explain what additional information you would need.

2a) (2 pts) What is $P(A, B)$?

2b) (2 pts) What is $P(A \mid B^C)$?

Hint: Use the information provided to create a joint probability table for A and B .

Note: You can easily create a table in Markdown using this format (double click on this cell to see the syntax to create this table): `| | Col A Title|Col B Title| |—|—|—| |Row A Title|Entry 1 |other stuff | |Row B Title|`Latex works: $\int_1^2 x \, dx$ | more stuff |

Write up your full solution to both questions in the SAME box below using LaTeX (not code). Show all steps fully justifying your answers.

	A	A^C	Total
B	0.2	0.2	0.4
B^C	0.1	0.5	0.6
Total	0.3	0.7	1

Given the provided numbers, it is possible to use the fact that the probabilities must add up to 1 to fill in the rest of the table.

2a) A and B , so 0.2.

2b) Given B^C , treat the second row on its own. Normalize by dividing each column by the total, 0.6. This yields $\frac{1}{6}$ for the column where A occurs.

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0.2 Question 4a: Poker! (4 pts)

A common example for discrete counting and probability questions are poker hands. Consider using a standard 52-card playing deck, with card ranks $[A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K]$ across the standard 4 suits: $[C, D, H, S]$.

Suppose we draw 5 cards at random from the deck without replacement.

In Poker, “**Three of a Kind**” is defined as a hand that contains **three cards of one rank and two cards of two other ranks**. Notice that in this definition, a **Full House** (a hand that contains three cards of one rank and two cards of another rank) is **NOT** classified as “three of a kind”. https://en.wikipedia.org/wiki/List_of_poker_hands#Three_of_a_kind

What is the probability of drawing 5 cards (without replacement) that are “three of a kind?”

Typeset your work using LaTeX below. Show work justifying all steps. You may leave your answer in terms of a ratio of products, but you should simplify away any combinatoric notation such as $\binom{n}{k}$ or $P(n, k)$.

First, find the number of three of a kind hands. The components of a three of a kind are picking the rank of the triplet, selecting 3 cards of that rank from the 4 different suits, pick 2 ranks from the remaining 12, then pick a suit for each of them. This can be expressed like so:

$$13 \cdot \binom{4}{3} \cdot \binom{12}{2} \cdot 4 \cdot 4 = 54,912 \quad (1)$$

This value can then be divided by the total number of 5 card hands

$$\frac{54,912}{\binom{52}{5}} = \frac{54,912}{2,598,960} \approx 2.9\% \quad (2)$$

0.3 Question 4biii (4 pts)

Write code in the space below that completes the following steps:

Step 1: Write a function to simulate 10,000 random draws from `cards` of 5 cards each, and check if each draw is Three of a Kind. The function should return the overall proportion of random hands (out of the 10,000) in which Three of a Kind was observed. If you have coded your simulation correctly, your answer to this part should be very close to your theoretical answer from Part 4a.

Step 2: Let's visualize how this simulation converges to the theoretical probability. In the video in Question 3, we plotted a running estimate of the probability of an event as a function of the number of trials in our simulation. Write code that completes 10,000 random draws of 5 cards each, but this time outputs a plot of a running estimate of the proportion of hands that are Three of a Kind as a function of the number of trials (from 1 to 10,000) in your simulation. **Include a red horizontal line on your plot with the theoretical probability that you calculated in part 4a.** Be sure to include a title on your plot and be sure to label both your axes on the plot, as well as include a legend.

```
In [46]: def probability_of_three_of_kind(deck: list[str], num_sim: int):
        three = np.array([three_kind(rng.choice(deck, size=5)) for i in range(num_sim)])
        three_running_frac = np.array([sum(three[:i+1]) / (i+1) for i in range(num_sim)])

        return sum(three) / num_sim, three_running_frac

        #probability_of_three_of_kind(cards, 10000)
        #4biii).Write your code for Step 1 above this line

In [47]: x = np.arange(1, 10001)
        prob, running_frac = probability_of_three_of_kind(deck, num_sim=len(x))

        print("Simulated probability", prob)

        fig, ax = plt.subplots(figsize=(12, 6))

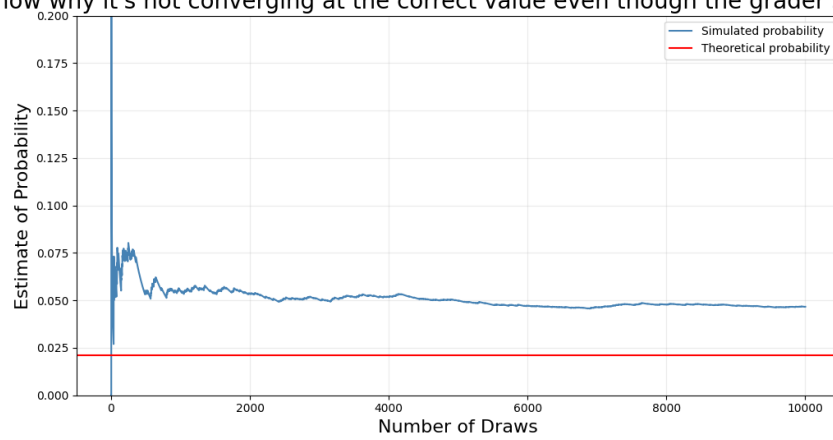
        ax.plot(x, running_frac, color="steelblue", label="Simulated probability")

        plt.axhline(y=54912/2598960, color='r', linestyle='-', label="Theoretical probability")

        ax.set_title("Running Estimate of Probability of a Three of a Kind\n(Don't know why it's not c
        ax.set_xlabel("Number of Draws", fontsize=16)
        ax.set_ylabel("Estimate of Probability", fontsize=16)
        ax.set_ylim(0, 0.2)
        ax.legend()
        ax.grid(True, alpha=0.25)
        # 4biii) Write your code for Step 2 above this line
```

Simulated probability 0.0467

Running Estimate of Probability of a Three of a Kind
(Don't know why it's not converging at the correct value even though the grader said it was fine)



0.4 Question 5a (4 pts)

5a i). What is the **mathematical definition** of the conditional probability of an event A given an event B? Give your answer using LaTeX and correct mathematical notation. You can assume $P(A) > 0$ and $P(B) > 0$

5a ii). You flip a fair coin twice. What is the probability that both flips are heads given that at least one of the flips is a heads? Explain/justify using math.

Enter your answer for part 5a i) in this cell (double click on this cell and write all steps using Markdown and LaTeX):

$$P(A | B) = \frac{P(A, B)}{P(B)} \quad (3)$$

Enter your answer for part 5a ii) in this cell (double click on this cell and write all steps using Markdown and LaTeX):

Since the coin is fair, let A be “both are heads” and B be “at least one of the flips is heads.” Therefore $P(A) = 0.25$ and $P(B) = 0.5 + 0.5 - 0.25 = 0.75$

$$P(A | B) = \frac{P(A, B)}{P(B)} \quad (4)$$

$$= \frac{0.25}{0.75} \quad (5)$$

$$= \frac{1}{3} \quad (6)$$

Therefore the probability is $\frac{1}{3}$

0.5 Question 5b (4 pts):

5b). Suppose a bag has 5 red marbles and 3 blue ones. We select 2 marbles without replacement (i.e. we select the first marble and then we don't put it back and then select a 2nd marble).

- i). What is the probability that the 2nd marble is red given that the first marble is red? Explain/justify using math.
- ii). What is the probability that both marbles are red? Explain/justify using math.

Enter your answer for part 5bi) in this cell (double click on this cell and write all steps using Markdown and LaTeX):

Let A be “the second marble is red” and B be “the first marble is red”

$$P(A | B) = \frac{P(A, B)}{P(B)} \quad (7)$$

$$= \frac{\frac{5}{8} \cdot \frac{4}{7}}{\frac{5}{8}} \quad (8)$$

$$= \frac{4}{7} \quad (9)$$

Therefore, if the first marble is red, then the second one has about a 57% chance of also being red.

Enter your answer for part 5bii) in this cell (double click on this cell and write all steps using Markdown and LaTeX):

Using the values from the previous part:

$$P(A, B) = P(A | B) \cdot P(B) \quad (10)$$

$$= \frac{4}{7} \cdot \frac{5}{8} \quad (11)$$

$$= \frac{20}{56} = \frac{5}{14} \quad (12)$$

Therefore there is about a 56% chance that both marbles are red.

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0.6 Question 6 (7 pts)

The accuracy of a diagnostic test is often described using the following terms:

- Test Sensitivity: Ability to detect a positive case (i.e. probability that the test is positive given that the person actually has the virus).
- Test Specificity: Ability to determine a negative case (i.e. the probability that a person tests negative given that they don't have the virus)).

Suppose a diagnostic test for a virus is reported to have 85% sensitivity and 95% specificity.

Suppose 3% of the population has the virus in question.

Answer the following questions all in ONE cell below using LaTeX. Show all steps.

6a) (6 pts). If a person is chosen at random from the population and the diagnostic test indicates that they have the virus, what is the conditional probability that they do, in fact, have the virus? Write up your full solution using LaTeX.

6b) (1 pt). Terminology: What is the **prior** and what is the **likelihood** in this scenario?

Let T be a positive test and V be actually having the virus.

$$\text{Sensitivity} = 0.85 = \frac{P(T, V)}{0.03} \quad (13)$$

$$0.0255 = P(T, V) \quad (14)$$

$$\text{Specificity} = 0.95 = \frac{P(T^C, V^C)}{0.97} \quad (15)$$

$$0.9215 = P(T^C, V^C) \quad (16)$$

$$(17)$$

	T	T^C	Total
V	0.0255	0.0045	0.03
V^C	0.0485	0.9215	0.97
Total	0.074	0.926	1

6a)

Find $P(V \mid T)$:

$$P(V \mid T) = \frac{P(V, T)}{P(T)} \quad (18)$$

$$= \frac{0.0255}{0.074} \quad (19)$$

$$= \frac{51}{148} \approx 34\% \quad (20)$$

Therefore the probability is about 34%.

6b)

The prior is the initial probability that the test yields a positive: 7.4%.

The likelihood is the probability that the test is positive if the person is sick (a.k.a. $P(T \mid V)$), which, using a similar process to part 6a, is 85%.

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0.7 Question 7a (3 pts)

To play a game, you have a bag containing 30 fair **four-sided dice**, with faces $\{1, 2, 3, 4\}$. This bag also contains 9 fair six-sided dice (faces $\{1, 2, 3, 4, 5, 6\}$) and 3 fair twenty-sided dice (faces $\{1, 2, 3, 4, \dots, 19, 20\}$). Call these 3 classes of die “Four”, “Six” and “Twenty” (or D_4 , D_6 , and D_{20} , for short). You grab one die at random from the box.

Work the following problems by hand and write up your full solution using LaTeX unless otherwise stated (but don't be afraid to simulate to check your result!).

You grab one die at random from the box and roll it one time. What is the probability of the event R_2 , that you roll a 2? Explain your reasoning mathematically (using LaTeX).

There is a $\frac{1}{4} \cdot \frac{30}{42}$ chance of rolling a 2 from a D_4 . Likewise, for a D_6 it's $\frac{1}{6} \cdot \frac{9}{42}$, and for a D_{20} it's $\frac{1}{20} \cdot \frac{3}{42}$.

$$P(R_2) = \left(\frac{1}{4} \cdot \frac{30}{42}\right) + \left(\frac{1}{6} \cdot \frac{9}{42}\right) + \left(\frac{1}{20} \cdot \frac{3}{42}\right) \quad (21)$$

$$= \frac{61}{280} \approx 22\% \quad (22)$$

Therefore the probability of R_2 is about 22%.

0.8 Question 7b (3 pts)

You grab one die at random from the box and roll it one time. Suppose you roll a 2. Given this information, what is the probability that the die you chose from the box is a Twenty-sided die? Write up your full solution using LaTeX. Show all steps.

Using the information from the previous part, $P(R_2) = \frac{61}{280}$ and $P(D_{20}, R_2) = \frac{1}{20} \cdot \frac{3}{42} = \frac{1}{280}$.

Find $P(D_{20} \mid R_2)$:

$$P(D_{20} \mid R_2) = \frac{P(D_{20}, R_2)}{P(R_2)} \tag{23}$$

$$= \frac{\frac{1}{280}}{\frac{61}{280}} \tag{24}$$

$$= \frac{1}{61} \approx 1.6\% \tag{25}$$

Therefore the probability that the die was a D_{20} is about 1.6%.

